The best discovery is that transmission temperature, current gear and torque-converter lockup may all be accessible through the transmission ECU using Toyota's proprietary **Mode 21** diagnostics. You shouldn't need to sniff the CAN bus to try them.

1. The four essential readings

Parameter	OBD request	ECU header	Confidence
Transmission fluid temperature	`21D9`	`7E1`	High — tested on V8 diesel LC200s from 2010 and 2013
Current gear	`21DA`	`7E1`	Experimental — documented on related Toyota transmissions
Torque converter lockup	`21DA`	`7E1`	Experimental — listed for 2008–2015 Land Cruisers
Boost / manifold pressure	`010B`	Automatic	Standard OBD2, if supported by your ECU

ScanGauge Australia publishes `21D9` and `21DA` for the 2008–2015 Land Cruiser family. Separately, an LC200 owner confirmed that `21D9` reads both transmission oil-pan and converter-outlet temperatures on V8 turbo-diesel Land Cruisers. ScanGauge A... +2

Important distinction: The temperature PID has strong model-specific evidence. For gear and lockup, I found working implementations on related Toyota vehicles, but not a verified test on a 2008 VDJ200 specifically. I'll treat those two as test candidates rather than guaranteed working codes.

2. Exact PID definitions and decoding formulas

A. Transmission fluid temperature

Best verified

The `21D9` request provides two useful temperatures from a single ECU response.

Reading	Formula	Units
ATF oil-pan temperature	`E-40`	°C
Torque-converter outlet temperature	`G-40`	°C

Both use request `21D9`, header `7E1`.

The simpler equations are accurate to approximately 1°C according to testing on the LC200. If you want greater precision, the published pan-temperature equation is:

```
(((((E*256+F)*0.07-400)/10)-32)*5/9)
```

Replace `E,F` with `G,H` for converter-outlet temperature. Torque Forum +1

For towing, I'd actually display both temperatures. The converter-outlet temperature can react faster to increased converter slip, whereas pan temperature is more representative of the bulk transmission oil temperature.

B. Actual transmission gear

The most promising older-Toyota request is:

Command	21DA Copy
Header	7E1

Torque Pro equation:

```
BIT(A:0)+2*BIT(A:1)+4*BIT(A:2)
```

Car Scanner equation:

```
GetBit(A,0)+2*GetBit(A,1)+4*GetBit(A,2)
```

These extract the three least-significant bits of response byte `A`, producing a numeric gear indication. The formula has been demonstrated on Toyota transmissions including the Prado A750, but needs verification on the VDJ200's six-speed transmission. pradopoint.c... +1

C. Torque-converter lockup

This conveniently uses the **same `21DA` request**.

Torque Pro equation:

```
BIT(A:4)
```

Car Scanner equation:

```
GetBit(A,4)
```

Expected output:

- `0` — lockup flag OFF
- `1` — lockup flag ON

This is the interpretation reported for older Toyota transmissions. It may represent an ECU lockup-control state rather than independently measured zero mechanical slip. pradopoint.c... +1

For example, if response byte `A` were `0x15`, the proposed decoder would report gear 5 and lockup ON. That's an illustrative value, not a captured VDJ200 response.

D. Boost pressure / MAP

Start with standard OBD2 data.

Reading	Request	Formula
Manifold absolute pressure	`010B`	`A` kPa, with scaling caveat below
Barometric pressure	`0133`	`A` kPa
Maximum MAP scaling information	`014F`	Response byte `D`

Boost pressure is calculated from absolute pressures:

$$\text{Boost (bar)} = \frac{\text{MAP} - \text{Barometric}}{100}$$

For example, MAP of 220 kPa and atmospheric pressure of 100 kPa corresponds to 1.20 bar boost.

One important Toyota-specific complication: Some Toyota diesel ECUs use an extended 320 kPa MAP range while still returning an 8-bit Mode 01 value. This has been documented on 1KD and 1GD diesel engines, but I haven't confirmed whether your 2008 1VD uses this scaling.

pradopoint.c... +1

If `014F` is supported and byte `D` is nonzero, the correct MAP calculation is:

$$\mathrm{MAP}_{kPa} = A_{010B} \times \frac{10 D_{014F}}{255}$$

This comes from the OBD diagnostic scaling standard. Studylib

An easy check is to switch on the ignition without starting the engine. MAP should approximately equal atmospheric pressure. If it is around 80 kPa when barometric pressure is around 100 kPa, scaling is probably the problem.

3. Using your Bluetooth 5.1 ELM327 adapter

I'd start with Car Scanner ELM OBD2 on Android or iOS. It supports custom requests, Bluetooth Low Energy adapters and bit-level formulas. Car Scanner ... +1

The main configuration is:

Custom sensor	Request	Header	Car Scanner formula
ATF Pan	`21D9`	`7E1`	`E-40`
ATF Outlet	`21D9`	`7E1`	`G-40`
Current Gear	`21DA`	`7E1`	`And(A,7)`
TC Locked	`21DA`	`7E1`	`GetBit(A,4)`

`And(A,7)` is just a shorter equivalent of extracting bits 0, 1 and 2.

For boost, first try the app's built-in MAP, barometric and turbo-boost readings. Add a custom formula only if it appears incorrectly scaled.

Bluetooth 5.1 is not necessarily Bluetooth LE: check whether your particular dongle uses BLE, Classic Bluetooth or both. On iOS, a standard Classic-only ELM327 normally won't work. Also, inexpensive ELM clones sometimes fail on longer multi-frame requests like `21D9`, even when standard engine parameters work. Car Scanner ... +1

4. Testing the actual ELM327 commands directly

If you prefer developing your own display or interrogating the ECU from a laptop, you can send commands to the ELM adapter directly.

These are read-only diagnostic requests.

Initialize the ELM327

```
ATZ
ATE0
ATL0
ATS1
ATH1
ATSP0
010C
ATDP
```

`ATSP0` auto-detects the vehicle protocol. `010C` establishes communication by requesting engine RPM, and `ATDP` reports the selected protocol.

On the usual ISO 15765-4 CAN 11-bit/500 kbps connection, the next commands can be:

Transmission ECU

```
ATSP6
ATCAF1
ATCFC1
ATSH7E1
21D9
21DA
```

`ATSH7E1` directs the requests to the transmission diagnostic address. Positive replies should contain `61 D9` and `61 DA` respectively, potentially across multiple CAN frames.

Engine ECU — pressure readings

```
ATSH7DF
010B
0133
014F
```

For a successful MAP response, expect `41 0B` followed by data. If you want to address just the engine ECU, you can also test `ATSH7E0`.

Every command is sent with a carriage return (`\r`), and you should finish reading the response, normally indicated by `>`, before sending the next command.

If the transmission requests return `NO DATA`, check the ECU header, protocol and ISO-TP multi-frame handling before concluding that the PID is unavailable. In particular, some clone adapters mishandle automatic flow control.

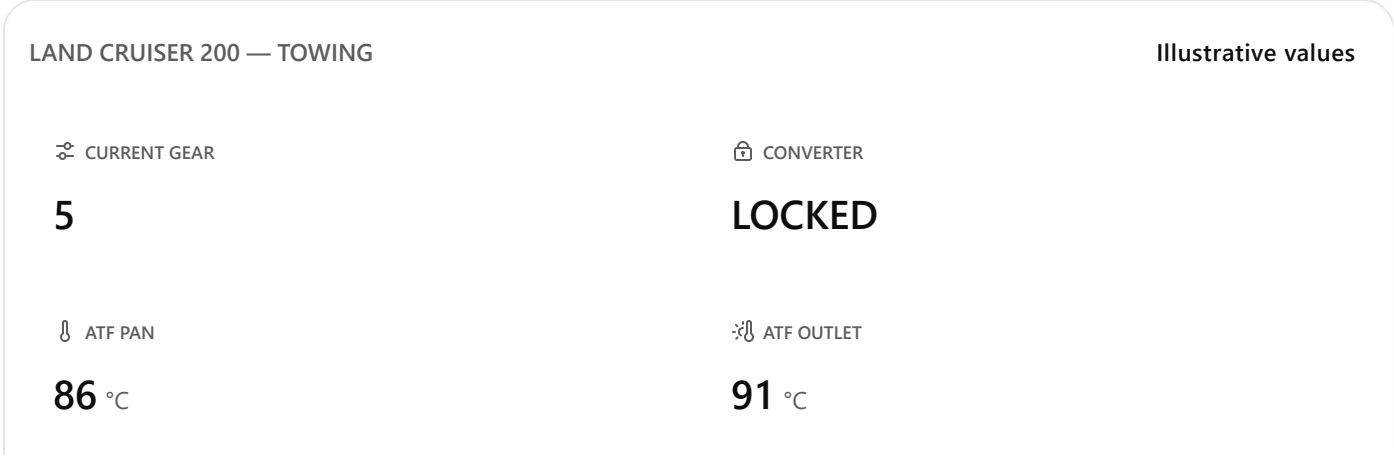
5. How to validate them on your vehicle

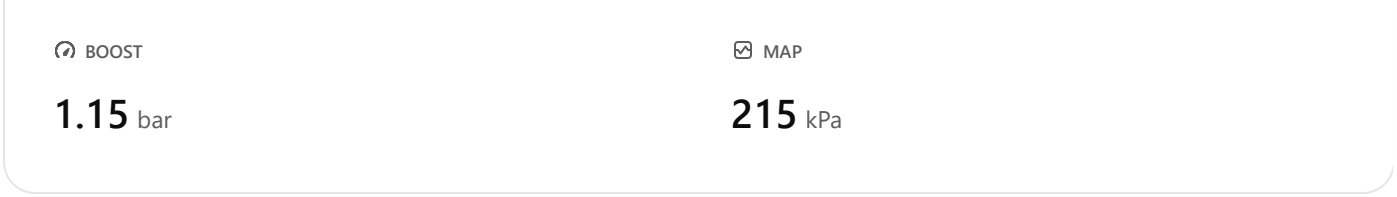
I'd do the first tests in this order.

1. **ATF temperature:** With the engine cold, compare the two temperatures against ambient temperature. Both should rise smoothly as the transmission warms up.
2. **Current gear:** During a controlled road test, compare the reported gear against manual gear selection and observable shifts. Verify all six ratios.
3. **Lockup:** On a warm transmission, cruise at a stable speed and watch the flag while gently varying the throttle. Cross-check it against the engine-speed response and ATF outlet temperature.
4. **MAP and boost:** Check atmospheric pressure with the engine stopped, then verify that measured manifold pressure rises under load.

Do the moving tests with a passenger operating the app or with data logging rather than watching the screen while driving.

6. My recommended dashboard





The readings shown above are mock values. For an actual installation, I'd poll gear/lockup and boost more frequently than ATF temperatures, and make the temperature gauges change appearance when approaching a warning level.

7. One correction to my earlier recommendation

The `221620` (lockup) and `221621` (gear) requests that I mentioned earlier are documented for newer Toyota vehicles using header `700`, including a 2021 Prado. They are **not verified for a 2008 VDJ200**. NewHilux

For your vehicle, `21DA` at `7E1` is the more appropriate initial test, although it remains unverified on that exact ECU.

The result of this research is encouraging: you likely need only three periodically polled diagnostic requests (`21D9`, `21DA` and `010B`) to obtain your essential four readings plus the additional converter-outlet temperature.

The remaining unknown is whether your 2008 VDJ200 responds to `21DA` with the same byte layout as the related Toyotas. Once you capture a couple of raw `21DA` replies, we can verify that conclusively and adapt the decoder if necessary.

If you want, I can:

- Request a step-by-step test plan for verifying the four essential PIDs via ELM327 Bluetooth
- Generate example raw OBD2 responses and decoding for gear and lockup status PIDs
- Advise on configuring Car Scanner app to read and decode custom PIDs on LC200